1. Personal details

- **Full name:** Robel K. Gebre, Ing, PhD
- **Address**: 4320 22nd Ave NW, Rochester Minnesota, 55901
- **Phone**: +1 507 779 2353
- **ORCID**: https://orcid.org/0000-0002-5746-0994
- **Google Scholar:** Link
- **Website:**

2. Employment details:

- **Name:** Mayo Clinic
- **Address:** 200 1st St SW, Rochester Minnesota, 55905
- **Date of employment:** 09/13/2021
- **Job Description:**
  - **Responsibilities:** Postdoctoral Research Fellow/Research Associate
  - **Completed projects:** Magnetic resonance imaging (MRI) scanner harmonization, novel tau positron emission tomography (PET) quantification, cognitive trajectory modeling from blood plasma biomarkers, Alzheimer's disease (AD) genetics using single nucleotide polymorphisms (SNPs) and machine learning, brain age prediction from MRI, and multiple system atrophy (MSA) signatures from structural and diffusion MRI.
  - **Active projects:** Early-onset prediction of MSA, clinical trial enrichment for AD anti-amyloid therapies, AD disease progression modeling, and cardiovascular disease (CVD) and cerebral amyloid angiopathy (CAA) profiling for anti-amyloid treatment (AAT).
  - **Accomplishments:** Novel tau PET metric, novel MSA imaging signatures, therapeutic window for AAT and clinical trials.

3. Degrees

- **Doctor of Philosophy, Medicine (PhD)**
  - *Date awarded*: 01 July 2021
  - *Institution*: University of Oulu, Oulu, Finland
  - *Major/Program*: **Medical Physics and Technology**
- **Master of Science (MSc. Ing.)**
  - *Date awarded*: 2016
  - *Institutions*: Joint Master's Degree Program at Czech Technical University (Prague, Czech Republic) and University of Groningen (Groningen, The Netherlands)
  - *Major/Program*: **Biomedical Engineering and Clinical Technology**
- **Bachelor of Science (BSc)**
  - *Date awarded*: 2013
  - *Institution*: Jimma University Institute of Technology, Jimma, Ethiopia
  - *Major/Program*: **Biomedical Engineering**

4. Awards and honours

- The Don Summers Memorial MSA Travel Fellowship Award (2025)

- Alavi-Mandell Award (2025)
- Young Investigator Travel Scholarship, Human Amyloid Imaging (2023)
- Mayo Clinic Radiology Research Fellowship, Mayo Clinic (2021)
- Newspaper covers on PhD thesis, Teknikka&Talous, Finland (2021)
- Orthopedic Research Society, Travel scholarship award (2019)
- Common European Master's Course in Biomedical Engineering Scholarship, Erasmus Mundus (2013)
- *Summa cum laude*, Jimma University (2013)
    - Named best young scientist in Ethiopia for BSc project on "*Wireless Infant Incubators*"

5. Scientific and societal impact

- Invited talks:
    - University of Mississippi Medical Centre, The MIND Centre (2025)
    - Karolinska Institute Department of Neurobiology, Sweden (2024)
    - Mayo-Karolinska Institute Seminars (2022)
- PhD work was featured in two newspaper articles (2021):
    - Lonkkamurtumat vähentyvät mutta lantion murtumiset yleistyvät – taustalla luukato | Tekniikka&Talous
    - Dissertation: Machine learning methods succeeded in identifying fracture cases from controls and classifying hip osteoarthritis from CT images. | Kauppalehti

6. Professional memberships

- American Autonomic Society (AAS)
- AAIC ISTART

7. Publications (Selected)

1. **Gebre, R.K.,** Raghavan, S., De Tora, M.E.J. *et al* (2026). Precise disease heterogeneity and progression quantification in MSA and Parkinson's disease using machine learning. *Sci Rep* **16**, 10579. https://doi.org/10.1038/s41598-026-45949-5
2. Raghavan S, **Gebre RK**, Vemuri P. Chapter 8 - Artificial intelligence: Relevance and applications in dementia. In: Sharma CP, Thomas V, Nakhmani A, Smith RJ, eds. Artificial Intelligence and Brain-Computer Interfaces in Healthcare. Academic Press; 2026:143-153.
3. **Gebre RK**, et al. *Heterogeneity of MRI Biomarkers in Multiple System Atrophy and Parkinson's Disease Assessed Using Machine Learning*. Sep 2025. In Review. Available at SSRN: http://dx.doi.org/10.2139/ssrn.5403293.
4. **Gebre RK**, et al. *Advancing Tau PET quantification in Alzheimer disease with machine learning: introducing THETA, a novel tau summary measure.* Journal of Nuclear Medicine. 2024; DOI: https://doi.org/10.2967/jnumed.123.267273
5. **Gebre RK**, et al. *Can integration of Alzheimer's plasma biomarkers with MRI, cardiovascular, genetics, and lifestyle measures improve cognition prediction?* Brain Communications. 2024; DOI: https://doi.org/10.1093/braincomms/fcae300
6. **Gebre RK**, et al. *Cross–scanner harmonization methods for structural MRI may need further work: A comparison study.* NeuroImage. 2023; DOI: 10.1016/j.neuroimage.2023.119912

7. Gunter NB, **Gebre RK**, et al. *Machine learning models of polygenic risk for enhanced prediction of Alzheimer disease endophenotypes.* Neurology Genetics. 2024; DOI: **https://doi.org/10.1212/NXG.000000000020012**
8. Ramanan VK, **Gebre RK**, et al. *Genetic risk scores enhance the diagnostic value of plasma biomarkers of brain amyloidosis.* Brain. 2023; DOI: https://doi.org/10.1093/brain/awad196
9. Raghavan S, Przybelski SA, Lesnick TG, Fought AJ, Reid RI, **Gebre RK,** et al. *Vascular risk, gait, behavioral, and plasma indicators of VCID*. Alzheimer's & Dementia. 2024; DOI: 10.1002/alz.13540
10. Tian J, Raghavan S, Reid RI, Przybelski SA, Lesnick TG, **Gebre RK**, et al. *White matter degeneration pathways associated with tau deposition in Alzheimer disease*. Neurology. 2023; DOI: 10.1212/WNL.0000000000207250
11. **Gebre RK**, et al. *Detecting hip osteoarthritis on clinical CT: a deep learning application based on 2-D summation images derived from CT.* Osteoporosis International. 2022; DOI: 10.1007/s00198-021-06130-y
12. **Gebre RK**, et al. *Structural risk factors for low-energy acetabular fractures*. Bone. 2019; DOI: 10.1016/j.bone.2019.07.004
13. **Gebre RK**, et al*. Discrimination of low-energy acetabular fractures from controls using computed tomography-based bone characteristics*. Annals of Biomedical Engineering. 2021; DOI: 10.1007/s10439-020-02563-4
14. Bhuyan M, **Gebre RK**, Finnilä M, et al. (2021). Preparation of filter by alkali activation of blast furnace slag and its application for dye removal. Journal of Environmental Chemical Engineering. 10. 107051. 10.1016/j.jece.2021.107051.

8. Peer reviewed journals

- Artificial Intelligence in Medicine
- Insights into Imaging
- Nature Communication
- Neurobiology of Aging
- NeuroImage
- Osteoporosis International
- Bone
- PlosOne

9. Research grants awarded

- Active Grants
  - Investigating Resistance and Resilience Mechanisms in Alzheimer's Disease. Funded by National Institute on Aging. (FP00090851.02-PR08) – Collaborator
- Completed Grants
  - Radiology Research Fellowship. Funded by Mayo Clinic Radiology – Recipient

10. Novel methods and software developed

1. **Two-Way Epistatic Interactions**: "*A **Novel** Approach to Encode Two-Way Epistatic Interactions Between Single Nucleotide Polymorphisms*" arXiV, Cornell University (Jun 2023): DOI: https://doi.org/10.48550/arXiv.2306.09175

2. ***Transition Point Estimator (TPE)***: Alzheimer's Imaging Consortium (2024): "*Identifying transition points in AD biomarkers using machine learning: A supplement to the reliable worsening method*" GitHub: https://github.com/RobelGebre/TPE-for-SHAP

3. ***THETA***: A **Novel** Tau Heterogeneity Evaluation in Alzheimer's Disease. GitHub: https://github.com/RobelGebre/THETA